Watching Each Other: Partisan Poll-Watchers and the Vote Count

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Abstract

Party poll-watchers oversee vote counting in many democracies, but their coverage is often uneven and incomplete. I use handwritten official records of three close Peruvian presidential runoffs, recovering poll-watcher presence and count timing. Voters are assigned to classrooms alphabetically, so I compare classrooms within schools. A watcher without an opposing counterpart moves the margin toward their party by up to 0.4 percentage points. Within schools, parties tend to co-locate their poll-watchers; when both are present, the margin is unchanged, fewer ballots are discarded as null, and counting runs 3.3% longer, suggesting deliberation over disputed ballots rather than fraud.

JEL Codes: D72, K16

Free and fair elections are foundational to democratic accountability, allowing voters to discipline and replace politicians (Besley and Case, 1995; Ferraz and Finan, 2008). When citizens cannot trust that their votes are counted as cast, the legitimacy of elected governments suffers (Berman et al., 2019). Yet concerns about the legitimacy of election

results are widespread: in the most recent round of the World Values Survey,[1] half of the countries had more than a third of respondents indicate that votes are often not counted fairly.

This paper studies the people who watch the count. In Peru, as in many countries,[2] parties may place poll-watchers in any polling station (classroom). Poll-watchers observe the vote and the count and can contest any irregularities or poll-workers' rulings on individual ballots during the count. Watcher coverage depends on each party's capacity to recruit volunteers, with party watchers absent from many classrooms. In the three runoffs I study, between 46% and 63% of classrooms counted their votes with no party watcher present. I ask what watchers do to the count when they are present, and what their uneven presence implies for electoral outcomes.

Estimating the effect of poll-watchers on election outcomes is typically challenging because parties recruit more poll-watchers in places where they have more support. Within a school, however, voters are assigned to classrooms alphabetically, so classrooms in the same school serve comparable groups of voters. I compare classrooms within a school, controlling for each classroom's first-round result and for the presence of the other party's watcher. Under these assumptions, I identify the effects of the four watcher configurations.

I use data from official voting registries scanned and published by Peru's electoral institution (Oficina Nacional de Procesos Electorales, ONPE) for three presidential runoffs: 2011, 2016, and 2021. From these registries, I extract handwritten information about poll-watcher and poll-worker presence, party affiliation, national ID numbers, and names. I use the first digit of the national ID number as a proxy for age, and names to construct a proxy for gender. I also extract timestamps that I use to measure the duration of voting and ballot counting.

[1] Haerpfer et al. (2022)

[2] For example, in most US states, representatives are permitted to observe voting and counting.

I find that a poll-watcher present without the opposing party poll-watcher shifts the vote margin toward their party by up to 0.4 percentage points, both raising their own party's vote share and lowering their rival's. Importantly, when both are present, the effects cancel and the margin is indistinguishable from classrooms with no poll-watcher. I also find that parties tend to match each other within schools, so both are present in the same classroom far more often than independent placement would produce, even though symmetric effects suggest reallocating a watcher between a matched and an unwatched classroom would leave the expected margin unchanged.

The evidence is consistent with poll-watchers primarily operating through the legitimate contestation of ballots during the count. Unlike previous studies that typically focus on voting outcomes such as null votes and turnout, I can also observe the duration of voting and the duration of the count. When both parties' poll-watchers are present, the share of null votes falls by 0.08 percentage points and the duration of the count increases by 3.3%, consistent with longer deliberations over disputed ballots. By contrast, I find no significant effects on blank votes or turnout, providing little evidence that fraud or turnout buying drive the results. The effect on the duration of voting is also below half of one percent, consistent with no major interventions while polls are open. The results are robust to alternative samples, including restricting to classrooms that opened on time and to those staffed by the same poll-workers in both rounds.

This paper makes three contributions to the literature. First, I add to the literature on partisan poll-watchers[3] by introducing a novel measure of the counting process: the duration of ballot counting, constructed from timestamps in official voting registries. This measure provides a direct way to examine what happens during the count, complementing studies that typically focus on turnout or vote shares. I use this measure to

[3]For studies on non-partisan poll-watchers, see Buzin, Brondum and Robertson (2016), Asunka et al. (2019), Hyde (2007), Hyde (2010), Leeffers and Vicente (2019), Sjoberg (2012), Ichino and Schündeln (2012), and Sjoberg (2016).

document a legitimate channel through which partisan poll-watchers can affect electoral outcomes: the contestation of ballot validity during the count. Counting takes 3.3% longer when both parties are present, consistent with more deliberation over disputed ballots. This channel contrasts with the illegitimate mechanisms emphasized in previous work, such as removing rival ballots in Argentina (Casas, Díaz and Trindade, 2017) and enforcing vote buying in Mexico (Ascencio and Rueda, 2019), as well as discretion over voter eligibility exercised by polling-station officials in India (Neggers, 2018) and the United States (Atkeson et al., 2014).[4] In my setting, a single official ballot rules out removing rival ballots and a national ID leaves little room for discretion over eligibility. The finding that poll-watchers can affect the count through legitimate contestation suggests that their role may matter across a broader range of electoral settings.

Second, I show that parties match poll-watcher assignments within schools by co-locating their poll-watchers in the same classrooms. Ascencio and Rueda (2019) find that parties match overall coverage levels across schools, and I extend this by finding they choose to co-locate at the most local level, across classrooms within a school. This occurs even though matching gains them no more than spreading out would. One potential explanation is that parties hold biased beliefs about each other's effects and intentions, which would be particularly relevant in an increasingly polarized environment.

Third, I study partisan poll-watching in presidential runoffs decided by narrow margins where both parties have strong incentives to monitor and influence the count but limited capacity to do so. Close races are more likely to generate challenges to election results: runner-up parties reject outcomes even in elections of high quality, and rejection becomes more common as the margin narrows (Hernández-Huerta, 2020). Narrow-margin elections can arise from the structure of electoral systems, such as the

[4]This channel operates in contexts where multiple valid identity documents give officials discretion to question eligibility. In Peru, a single national ID leaves little room for such discretion.

U.S. Electoral College, as well as from how parties mobilize voters (Geruso and Spears, 2026; Levine and Martinelli, 2024). Additionally, having less capacity to manipulate results contrasts with Casas, Díaz and Trindade (2017) and Ascencio and Rueda (2019), who study elections with incumbents, multiple parties, and settings where the party with greater coverage had poll-watchers in around 90% of classrooms. With no incumbent in the race, neither side can rely on control of the state, but rather on its own organization to monitor the count.

Existing evidence suggests that voters place greater trust in elections when both parties are represented in the monitoring process. In survey experiments in the United States, respondents of both parties judge an election fairest when poll-watchers from both parties are present (Cohen and Sheagley, 2024), and in Malawi and Kenya, voters' confidence in tally sheets rises most when they are co-signed by agents from both parties (Mbozi, 2025). Combined with my finding that symmetric coverage does not shift the vote margin, this suggests that promoting coverage by both parties could strengthen confidence in the count without tilting the result toward either party.

The rest of the paper proceeds as follows. Section 1 describes the Peruvian election system, and Section 2 describes the data used. Section 3 presents the empirical strategy. Section 4 presents the main results and mechanisms, and Section 5 the robustness checks. Section 6 concludes.

1 Elections and Poll-Watchers in Peru

This section describes the institutional features of Peruvian elections relevant to this paper: the electoral rules, how votes are cast, and the role of party poll-watchers. The three elections studied were all closely contested runoffs. The 2011 runoff was won by fewer than half a million votes, a margin of 2.7%. The 2016 and 2021 elections were

razor-thin, both decided by fewer than 50,000 votes, corresponding to a margin of 0.2% of votes cast. Notably, the same party lost all three elections, while the winning party was different in each.[5]

1.1 Election Rules

In Peru, voting is compulsory,[6] and the presidency is generally decided in two rounds. In the first-round, voters choose among multiple presidential candidates: 11 parties competed in 2011, 14 in 2016, and 18 in 2021. If no candidate receives more than 50% of the vote, the two leading candidates advance to a runoff. The second-round determines only the presidency, as congress has already been elected by that point.[7] Presidential reelection is not permitted in Peru, and given the high degree of party fragmentation, no incumbent party participated in any of the three runoffs studied in this paper.

1.2 How Votes are Cast: Schools and classrooms

Elections are held at electoral sites, which in Peru are typically schools. Each voter must vote within their district of residence, but may select their school ahead of the election, typically choosing one near their home.[8] Within each school, voters are assigned to a polling station (a classroom) by the electoral institution (ONPE), in alphabetical order. Each classroom generally serves between 150 and 300 voters and is presided over by one poll-worker, assisted by two others, all drawn from the registered voters of

[5]Fuerza Popular went on to win the 2026 runoff by 0.26 percentage points. The 2026 election also shows how logistical failures can feed fraud narratives, particularly in the first-round, when partisan poll-watchers are almost absent. The late arrival of voting materials delayed the opening of many classrooms in Lima, voting was extended by a day in some locations, and the head of ONPE resigned before the runoff. Some parties alleged fraud, but the OAS and European Union observation missions found no evidence of it.

[6]Fines for not voting are between 5 USD and 25 USD.

[7]Because there are no strong parties in Peru, votes are highly disperse and hence, it is often the case that no candidate has won a majority in congress before heading to the second-round.

[8]Around 10% of voters choose their own school; the rest are assigned to a school within their district by the electoral institution.

that classroom and assigned by ONPE ahead of the election. These poll-workers are responsible for setting up and reviewing election materials, verifying voter IDs, and conducting the count. Upon entering the room, a voter approaches the poll-workers, presents their national ID, and receives a single official ballot supplied by ONPE and listing all candidates. The voter then marks the ballot behind a private booth and deposits it in a ballot box.[9] Additionally, parties may assign poll-watchers to monitor the process and observe and contest the count, which is the focus of this paper.

1.3 Party Poll-Watchers

Each party may assign poll-watchers to any classroom to monitor the voting process and the vote count at the end of the day. Because parties generally do not have enough poll-watchers to cover all classrooms, they must allocate them selectively, leaving some classrooms unmonitored. My identification strategy relies on this allocation being conditionally random within a school, which I discuss in Section 3. During the vote, poll-watchers are seated alongside the poll-workers inside the classroom. Figure B.1 illustrates this setup, showing the exterior of a school, voters lined up inside a classroom, and the key actors present during the vote: poll-workers, a voter at the booth, and party poll-watchers.

Poll-watchers have an official role before, during, and after the election. Before voting begins, they verify that the voting booth is properly set up and that the classroom has all the necessary ballots. During voting, they may verify that the voting booth has no marks and raise observations if irregularities arise. After voting closes, they ensure that the total number of ballots matches the total number of voters, and may contest poll-workers' rulings on individual ballots.

[9] Peru uses the "Australian ballot" system, also used in the United States. Unlike the party-supplied ballots used in the Argentine elections studied by Casas, Díaz and Trindade (2017), this system rules out removing ballots selectively by party.

If poll-watchers affect the count, their presence should raise their own party's vote share and lower their rival's, widening the margin between the two. There is a legitimate way they have to do this: contesting poll-workers' rulings on whether individual ballots are valid.[10] Voters make a range of mistakes when marking their ballots (Dee, 2007). In Peru these mistakes surface as ballots whose validity the poll-workers must judge. During the count, poll-watchers may argue that a ballot for their party was wrongly invalidated, increasing their party's vote count if accepted, or that a ballot counted for a rival should be invalidated, decreasing the rival's vote count if accepted.[11] Poll-workers may be more receptive to arguments for counting a vote than for discarding one, since the latter can look like suppressing a voter's choice. If so, poll-watchers should have a larger effect on their own party's share than on their rival's. Two further implications follow. When both parties' poll-watchers are present, both vote shares could rise as fewer ballots are deemed null, while maintaining the same vote margin. At the same time, opposing challenges and counterarguments may prolong deliberation over disputed ballots, slowing the count.

Prior work has also documented illegitimate channels, but Peru's institutions make most of them unlikely here. In Argentina, each party prints and distributes its own ballots, and Casas, Díaz and Trindade (2017) argue that poll-watchers affect results mainly by removing opposition ballots; in this setting, as in many other countries, a single official ballot lists all candidates. In India, poll-workers exercise discretion over voter eligibility in areas with low identity document coverage, which Neggers (2018) finds to be a source of electoral bias; in Peru, voters must present a national ID, which leaves little room for such discretion. In Mexico, Ascencio and Rueda (2019) find that reading voters' names aloud as they approach the classroom could be used by

[10] It is unclear whether poll-watchers are correcting errors or introducing a pro-party bias that deviates from the intended vote.

[11] For example, a cross over the candidate's picture rather than the party logo is also valid, whereas a circle instead of a cross is invalid.

poll-watchers as a tool to irregularly control voting behavior by facilitating turnout buying enforcement; in Peru, poll-workers are not instructed to read names out loud. However, poll-watchers could still intimidate voters or otherwise disrupt voting while polls are open, which would lower turnout or lengthen the voting process. Finally, poll-watchers could facilitate or deter outright fraud, such as filling in blank ballots or those of absent voters, which would appear as higher turnout or fewer blank votes. I test for these traces in Section 4.

2 Data

The data comes from two official sources from ONPE, the Peruvian electoral institution.[12] The first source is the official vote counts, which ONPE publishes publicly at the classroom level for both rounds of each election. These provide vote totals by party, blank votes, null votes, and turnout for each of the classrooms. The number of schools, classrooms per school, and voters per classroom varies by year as shown in the first panel of Table 1. The second source is ONPE's official voting registries, one per classroom, scanned and uploaded after each election. After the count is completed, the three poll-workers destroy the ballots and record the total votes received by each party on the registry. Any poll-watchers present also sign the sheet and write down their party affiliation.

This handwritten information is my primary source of data on poll-worker and poll-watcher presence and party affiliation. In addition, both poll-workers and poll-watchers are required to write down their national ID number and full name, which I use to construct proxies for age and gender. From these records, I also extract timestamps

[12] I have only considered votes by citizens within Peru, excluding those cast in a foreign country. This is due to difficulties in obtaining some of the information from foreign registries due to the differences in format.

that log the opening and closing times of both the voting period and the vote count, which I use to construct the duration of each for every classroom. Further details on the data collection and processing procedure are provided in Appendix A.

Table 1 reports descriptive statistics for all three elections studied in this paper, which were decided by narrow margins. The 2011 runoff was decided by fewer than 500,000 votes (2.7% of votes cast), while the 2016 and 2021 runoffs were each decided by fewer than 50,000 votes (0.2%). With narrow margins like these, even small differences in how votes are counted could plausibly determine the outcome.

Poll-watcher coverage is considerably lower than in comparable settings. Across the three elections, between 46% and 63% of classrooms had no poll-watcher present, meaning that in a large share of classrooms the vote count was conducted with no party supervision at all. [13] As I will show in the next section, the identification strategy relies on poll-watcher allocation being unrelated to potential outcomes within a school, after controlling for predicted second-round vote shares based on first-round results and the presence of the opposing party's poll-watcher.

3 Empirical Strategy

The main empirical challenge is that poll-watcher assignment is not random across the country. Because poll-watchers are volunteers, parties find it easier to recruit them in areas where they have more support. Panel (a) of Figure 1 plots poll-watcher presence against party support at the district level and shows this relationship clearly. A simple comparison across schools would therefore confound the effect of poll-watchers with differences in underlying electoral preferences.

[13] This stands in contrast to settings like Argentina (Casas, Díaz and Trindade, 2017) and Mexico (Ascencio and Rueda, 2019), where coverage is substantially higher. In both countries, the party with greater coverage had poll-watchers present in around 90% of classrooms.

Within a school, however, voters are assigned to classrooms in alphabetical order, which produces groups of voters with similar ideological preferences across classrooms within the same school. Differences in poll-watcher presence across these classrooms therefore provide a source of within-school variation that is less likely to reflect differences in electoral preferences.[14] I use this variation to estimate the effect of poll-watchers on the vote margin using the following specification:

$$VoteMargin_{sc} = \alpha_s + \delta_W Obs_{sc}^{W} + \delta_L Obs_{sc}^{L} + \delta_{both} Obs_{sc}^{both} + \beta X_{sc} + \epsilon_{sc} \quad (1)$$

Where $VoteMargin_{sc}$[15] is the vote share margin between the two parties in classroom c of school s. I label the eventual runoff winner as W and the runner-up as L throughout. Obs_{sc}^{W}, Obs_{sc}^{L}, and Obs_{sc}^{both} are indicators for whether a poll-watcher from the winning party alone, the losing party alone, or both parties are present in the classroom. α_s denotes school fixed effects and X_{sc} includes controls for other variables that may be related to poll-watcher assignment, discussed below. I expect δ_W to be positive, δ_L to be negative, and δ_{both} to be smaller in magnitude than either, as the two poll-watchers would offset each other to some extent.

The key assumption is that poll-watcher presence is unrelated to potential outcomes within a school. Conversations with electoral officials and party coordinators on election day provide initial anecdotal support as they suggest that within-school assignments follow no sophisticated or nationally coordinated strategy.

Below, I discuss potential threats to identification and how I address them. First, alphabetical assignment does not guarantee that classrooms within a school are perfectly

[14] Cantú (2014) and Casas, Díaz and Trindade (2017) find evidence consistent with this in the Mexican and Argentinian contexts respectively

[15] Defined as the winner's vote share minus the loser's vote share in percentage points.

homogeneous. Parties could observe differences across classrooms that, if correlated with poll-watcher assignment, would introduce bias. Panel (c) of Figure 1 plots vote shares and poll-watcher presence by the first initial of the lead poll-worker's surname and shows no systematic variation across initials, consistent with alphabetical assignment producing groups of voters with similar underlying preferences among voters who already live in the same district. However, because classrooms have the same group of voters in both the first and second-round, parties could in principle use first-round results to inform their poll-watcher assignments within a school. Panel (b) of Figure 1 shows that even within a school, first-round vote shares are a strong predictor of second-round vote shares. If parties exploited this and assigned their poll-watchers to classrooms where they expected to perform better, estimates would be biased away from zero.

I address this by controlling for the predicted second-round vote margin, $\widehat{VoteMargin}_{sc}$, in all specifications. For each runoff candidate $p \in \{W, L\}$, I regress the candidate's second-round vote share on the first-round vote shares of all parties, with school fixed effects, separately for each election:

$$VoteShare^{p}_{sc} = \mu^{p}_{s} + \sum_{k=1}^{K} \gamma^{p}_{k} \, FirstRound^{k}_{sc} + u^{p}_{sc}, \qquad p \in \{W, L\}, \tag{2}$$

where $FirstRound^{k}_{sc}$ is the first-round vote share of party k in classroom c of school s. The predicted margin is the difference between the two fitted values, including the estimated school fixed effects: $\widehat{VoteMargin}_{sc} = \widehat{VoteShare}^{W}_{sc} - \widehat{VoteShare}^{L}_{sc}$. Because it is built from each classroom's own first-round results, this control captures the differences across classrooms within a school that show up in first-round voting, which are the ones parties could most plausibly observe and act on. As a further check, I use the predicted margin as a placebo outcome, estimating equation (1) with $\widehat{VoteMargin}_{sc}$ on the left-hand side and omitting it from the controls. Section 5.3 reports these estimates.

Second, even if no unobserved characteristics influenced poll-watcher assignment, poll-watchers may respond to the other party's presence, which they can observe on election day. An important advantage of the Peruvian setting is that there are only two parties in the runoff, which means I can fully account for all four possible combinations of poll-watcher presence: no poll-watcher (omitted), winner party alone, loser party alone, and both. This is less feasible in settings with more parties, such as Casas, Díaz and Trindade (2017) and Ascencio and Rueda (2019), where accounting for all possible interactions is less viable. By including all four cases explicitly in the specification, I control for this form of strategic co-location. In Section 4.3 I show that this is relevant, since both parties watch the same classroom far more often than would be expected from independent assignment. Omitted configurations may then be correlated with the included ones and carry effects of their own. With two candidates there is nothing left out.

Third, a potential source of measurement error in poll-watcher presence arises if party poll-watchers stepped in to replace absent poll-workers on election day. In this case, they would appear in the voting registries as poll-workers rather than poll-watchers, leading their presence to be undercounted. Because a poll-watcher acting as a poll-worker would provide unobserved partisan representation, this would likely bias my estimates toward zero. I assess this by restricting the sample to classrooms whose poll-workers were the same in both the first and second-round. Since poll-watcher presence is limited in the first round, such replacements are less likely to occur then. This restriction therefore helps exclude classrooms where poll-watchers replaced absent poll-workers in the second round. Section 5.1 reports these estimates. A related concern is that poll-workers themselves may have had undisclosed party affiliations, and that parties assigned poll-watchers accordingly. This is unlikely to be a major issue given that formal party affiliation is very low in Peru, at under 0.5% of the population according

to the 2021 household survey, and poll-workers are drawn at random from the registered voters of each classroom.

4 Results

4.1 The Effect of Poll-Watchers on the Vote Margin

Table 2 reports results from various specifications of equation (1). Columns (1) through (5) progressively add fixed effects and controls on a common sample, while column (6) reports the preferred specification on the full sample. In all specifications with school fixed effects, a party's poll-watcher present alone shifts the vote margin in that party's favor. In column (6), the margin moves 0.31 percentage points toward the winner when only the winner's poll-watcher is present, and 0.38 percentage points toward the loser when only the loser's is.

The first specification in column (1) regresses the vote margin on each poll-watcher presence category. Column (2) adds district fixed effects, which capture most of the variation and causes the largest drop in coefficients from 10 to around 1 percentage point. Column (3) adds school fixed effects, which causes the coefficients to drop further to 0.48 percentage points for the winner party poll-watcher, and to -0.31 percentage points for the loser party. This illustrates the selection problem when relying on across-school variation in poll-watcher presence: parties tend to have greater poll-watcher presence in schools where their support is higher.

Importantly, once school fixed effects are included and identification relies on within-school variation in poll-watcher presence, the coefficients remain stable as first-round controls and poll-worker demographics[16] are progressively added in columns (4) and (5). Since demographic controls are not always available and their inclusion does not affect

[16] indicators for gender and age, proxied from first names and national ID numbers.

the estimates, column (6) reports results for the full sample excluding the demographic controls from column (5). This serves as the preferred specification throughout the rest of the paper.

The stability of the estimates to the addition of first-round results and salient characteristics of classroom poll-workers indicates that the results are not driven by these observable sources of strategic assignment. It cannot, however, rule out selection on unobserved classroom characteristics, which I examine further in Section 5.

These estimates come from a setting with less room for manipulation than prior work has studied. Casas, Díaz and Trindade (2017) find that poll-watchers raise their party's vote count by 1.5% on average in Argentina, and by up to 6% for some parties, where each party prints and distributes its own ballots. In Peru, as in the United States and many other democracies, a single official ballot rules out this form of manipulation. Yet I still find that poll-watchers affect the vote count.

4.2 Mechanisms: Effects on the Vote Count

To understand what drives these effects, I examine the mechanisms discussed in Section 1.3, including legitimate challenges to ballot validity and several forms of electoral manipulation. Columns (2) through (8) of Table 3 test for the patterns each mechanism would be expected to produce.

Starting with the vote share decomposition in columns (2) and (3), the results are consistent with poll-watchers increasing their own party's vote share and reducing their rival's. When the winning party's poll-watcher is present alone, the winner's vote share increases by 0.16 percentage points and the loser's falls by a similar amount. The pattern is symmetric for the losing party's poll-watcher. However, in this case the effect on their vote share is somewhat larger than the effect on their rival's, which suggests that poll-watchers may be more effective at defending their own votes than at contesting

their opponent's, as discussed in Section 1.

When looking beyond party vote shares, the evidence points toward the legitimate channel as the primary mechanism. When both poll-watchers are present, the effect on the vote margin is not significant, but null votes fall by 0.08 percentage points. Importantly, this is reinforced by the fact that count duration increases by 1.3 minutes, which represents a 3.3% increase over the mean of 38.3 minutes[17], consistent with longer deliberations over disputed ballots during the count.[18] By contrast, the effects on blank votes and turnout are small and generally insignificant across most configurations, which argues against fraud or turnout buying as relevant mechanisms. The effects on voting duration are also small relative to the mean of 537.1 minutes, suggesting that poll-watcher presence does not meaningfully alter the voting process itself.

4.3 Poll-Watcher Placement

Poll-watchers are a scarce resource, and how a party spreads them across the classrooms of a single school is an allocation decision parties have to make. The registries record presence classroom by classroom, which lets me observe how they assign resources at the most local level. Within a school, parties do not choose classrooms independently of one another. Panels (a) and (b) of Figure 2 show this clearly: panel (a) plots the observed share of classrooms with both parties' poll-watchers against the share expected under independent assignment, and panel (b) compares the actual distribution to both the random and maximum poll-watcher match benchmarks. Both panels indicate that poll-watchers from both parties are considerably more likely to be present in the same

[17] Importantly, this effect remains remarkably stable across years, with both poll-watchers increasing vote count duration by approximately 3.1% in 2011, 3.3% in 2016, and 3.8% in 2021.

[18] Poll-watchers can monitor multiple tables if they have enough time. This could potentially introduce bias if poll-watchers, after completing the vote count at their assigned table, move to another table that is taking longer. However, the results are unchanged when I exclude classrooms with repeated poll-watchers given their observed ID.

classroom than would be expected under independent assignment. In that case, one would expect both to be present in around 23% of classrooms. In practice, both are present in 34% of classrooms, close to the theoretical maximum of 35% if parties always matched their opponent's poll-watcher presence, indicating that parties systematically place poll-watchers where the other party is also present. This goes beyond Ascencio and Rueda (2019), who find that parties tend to match the level of coverage at the school level. I find that parties match assignments within a school by co-locating their poll-watchers in the same classrooms.

One explanation is that parties match where they expect the opponent to be more effective if left alone, given that matching neutralizes the effect. However, both parties have a similar effect when alone: the two one-sided effects are statistically indistinguishable in magnitude ($p = 0.50$), so moving a watcher from a matched classroom to an unwatched one would leave the expected margin unchanged. It could be that this is simply reflecting that the parties have similar effects *on average*, but that may mask heterogeneity across classrooms. Ultimately, only one party needs to be advantaged by co-location rather than separation for that to be the equilibrium. A second possibility is that parties are not responding to each other at all, but ranking classrooms on another characteristic observable to them and not controlled here, which would produce the same co-location with neither reacting to the other. Column (5) of Table 2 controls for the demographic characteristics of poll-workers, with no significant change in the estimates, which makes that particular characteristic an unlikely confounder. It does not establish that parties ignore it, or some other feature I do not observe, when choosing classrooms.

A third possibility is that parties may hold biased beliefs about what their opponents would do if left unmonitored, which could also explain their preference for co-locating. In a polarized and distrustful environment, it is possible that even two parties who are

operating legitimately believe that the other may not, and therefore prefer to monitor each other's actions as closely as possible.

5 Robustness Checks

Results from the main specification are robust to a number of alternative samples and specifications, all of which are summarized in Figure 3.

5.1 Classrooms with the Same Poll-Workers as the First-Round

One concern is that poll-watchers may have replaced absent poll-workers on election day, appearing in the voting registries as poll-workers rather than poll-watchers. To address this, I identify these classrooms from the poll-worker IDs recorded in the voting registries. Matching IDs across rounds is imprecise and tends to produce false negatives rather than false positives, so this restricted sample is best understood as a conservative subset of classrooms where replacement is unlikely to have occurred. Results are similar to the main estimates, suggesting that unobserved poll-watcher replacement is not widespread enough to meaningfully affect the findings.

5.2 On-Time Classrooms

A related proxy for poll-worker replacement is whether a classroom opened on time. When all poll-workers show up as assigned, the classroom generally opens on schedule; delays are more likely when a poll-worker is absent and needs to be replaced. I therefore restrict the sample to classrooms that opened on time, using the opening timestamps recorded in the voting registries. Results are again similar to the main estimates, further

supporting the conclusion that unobserved poll-watcher replacement is not driving the findings.

5.3 Placebo Test: Predicted Second-Round Votes

Figure 3 also displays, as a placebo test, the effect of each poll-watcher configuration on the predicted second-round vote margin based on first-round results. These effects are not significantly different from zero across most configurations, supporting the identifying assumption that parties are not systematically assigning poll-watchers to classrooms where they expect to perform better. In some cases the placebo estimates are slightly positive, suggesting a mild tendency for parties to favor classrooms where the winning party had a projected advantage. However, the loser party's placebo coefficient has the opposite sign to its coefficient on the actual vote margin, and any remaining tendency is addressed by controlling for predicted second-round vote shares in all specifications.

6 Conclusion

I find that a poll-watcher present without a rival shifts the vote margin toward their party by up to 0.4 percentage points, raising their own party's vote share and lowering their rival's. When both parties' poll-watchers are present, their effects offset and the margin is indistinguishable from classrooms with no poll-watcher. Because the two parties largely match each other's placements, one-sided coverage is uncommon in the elections I study. If coverage resembled that of other countries from prior studies, one side alone would watch roughly half of all stations, enough to move the national margin by about 0.2 percentage points, close to the margins that decided the 2016 and 2021 runoffs.

The evidence is consistent with legitimate contestation of ballots during the count as the primary mechanism. Poll-watchers appear to be more effective at defending their own party's votes than at contesting their opponent's votes, consistent with poll-workers being more receptive to arguments in favor of counting a vote than to arguments for invalidating one. This is corroborated by the effect on count duration: when both poll-watchers are present, the duration of the ballot count increases by 3.3%, consistent with longer deliberations over disputed ballots, while the effects on blank votes, turnout, and voting duration are small and insignificant across most configurations.

These findings are relevant beyond the Peruvian context. Narrow-margin elections can arise from the structure of electoral systems, such as the U.S. Electoral College, as well as from how parties mobilize voters (Geruso and Spears, 2026; Levine and Martinelli, 2024). Balanced coverage neutralizes the effect on the margin and is also the arrangement voters trust most (Cohen and Sheagley, 2024; Mbozi, 2025). The results therefore help characterize the implications of different poll-watcher configurations for both electoral outcomes and perceptions of electoral legitimacy. At the same time, my findings offer some insight into how parties assign their resources at the most local level, but further research is needed to understand if their beliefs about opponents' behavior shape those decisions.

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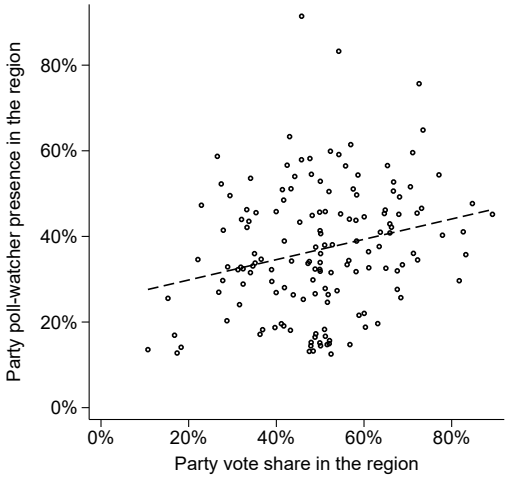
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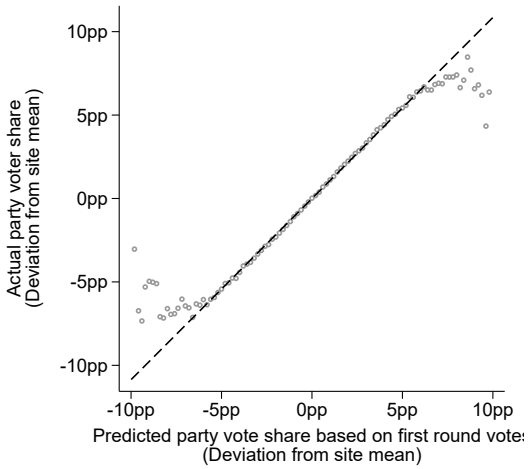
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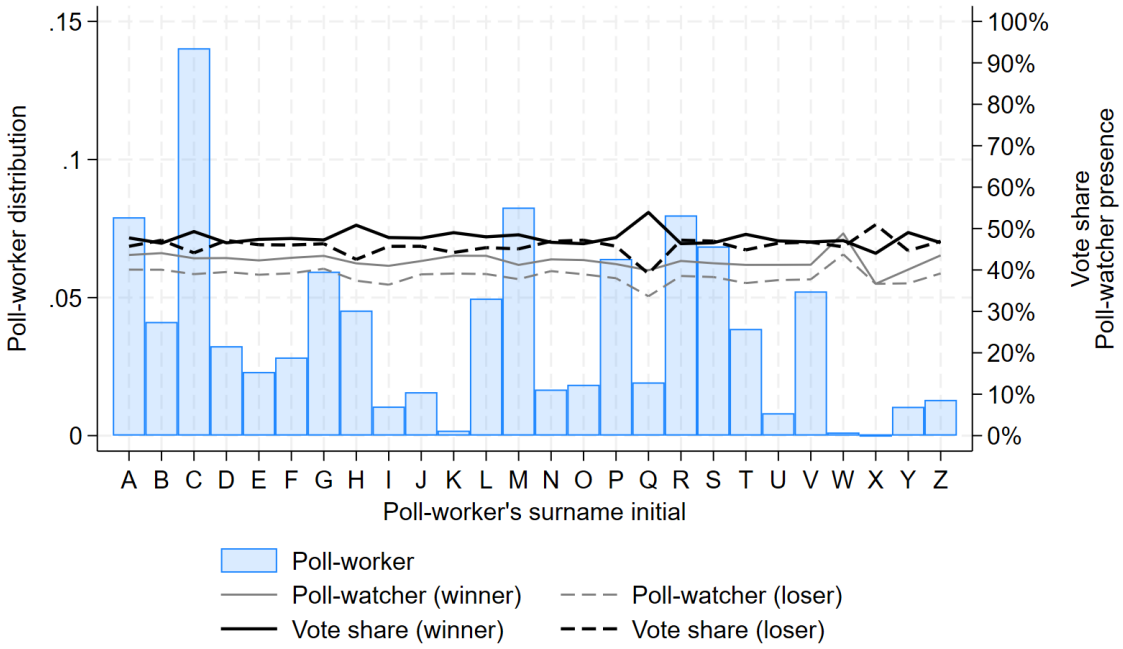
Figure 1: Identification strategy assumptions



(a) Vote share and observer presence



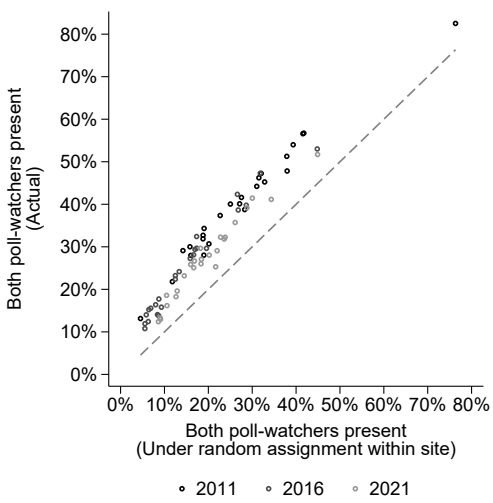
(b) Actual vs predicted second round votes



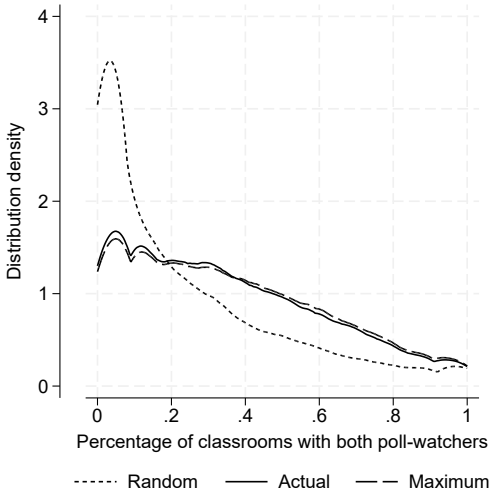
(c) Vote share and poll-watcher presence by last-name initial

Notes: Panel (a) shows the relationship between party support and poll-watcher presence. Panel (b) shows that even within a polling site, first round votes are a strong predictor of second round votes. Panel (c) shows, on the left axis, the distribution of poll workers and poll-watchers by first initial of the lead poll worker's last name and, on the right axis, the average vote share for the winner and runner-up parties. The x-axis shows the first initial of the lead poll worker's last name, which serves as a proxy for the last names of voters assigned to that classroom, since voters are assigned to classrooms alphabetically. The panel shows no strong association between alphabetical assignment and voting patterns, which supports the assumption that classrooms can be considered similar within a school. Moreover, since these are national averages, within-site variation would be even smaller. Main estimates are robust to excluding initials that exhibit the largest differences in vote shares.

Figure 2: Poll-watcher matching across classrooms



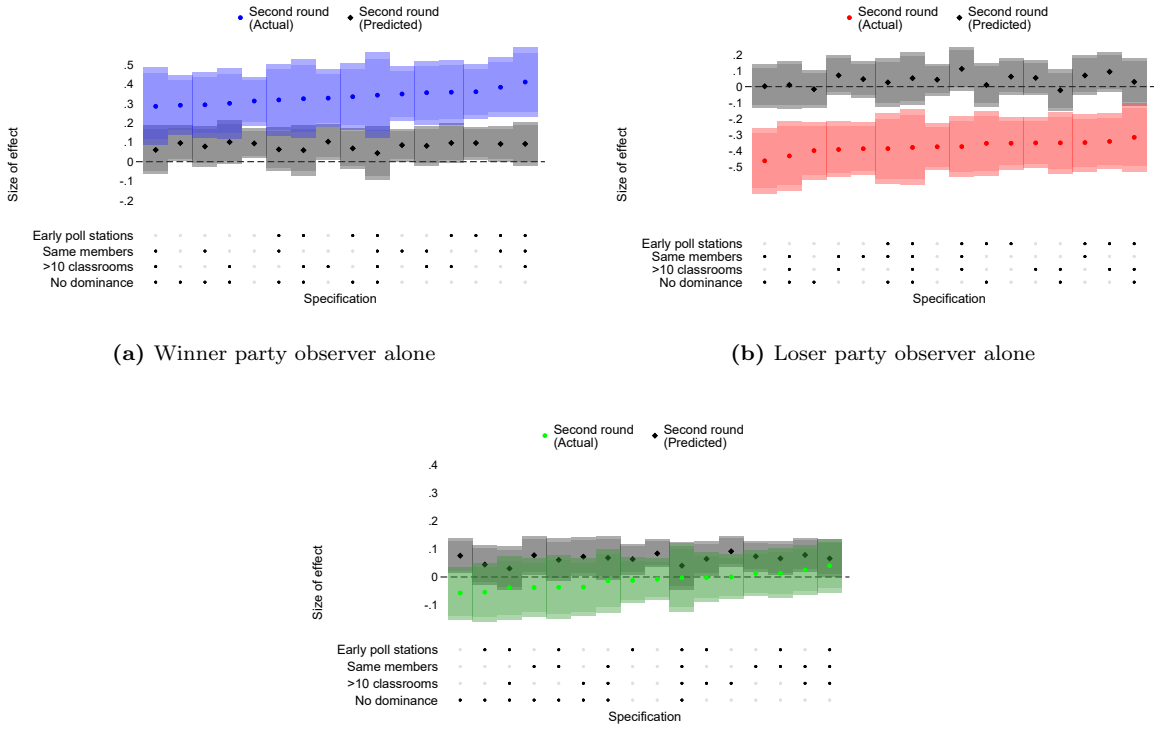
(a) Observers matching each others' presence



(b) Both poll-watchers' presence under different assumptions

Notes: Panel (a) plots, for each region-year, the share of polling stations with poll-watchers from both parties on the y-axis against the share that would be expected under random within-site assignment on the x-axis. Panel (b) shows the distribution of classrooms with poll-watchers from both parties under three assignment strategies: (i) random assignment by each party independently (short dash), (ii) the actual observed assignment (solid line), and (iii) the maximum possible overlap, if each party perfectly matched the other's poll-watcher placement (long dash). Together, the panels show that poll-watchers from both parties are considerably more likely to be present in the same classroom than would be expected under independent random assignment.

Figure 3: Effect of poll-Watchers



(a) Winner party observer alone

(b) Loser party observer alone

(c) Both observers

Notes: I show effects on the vote share margin (winner minus loser) of different observer assignments: winner party alone in panel (a), shown in blue; loser party alone in panel (b), shown in red; and both observers in panel (c), shown in green. In each case I show results on the predicted vote share margin based on first round results in grey. I do this for different samples: tables that open on time, with more than 10 classrooms, with the same set of table members than in the first round and without areas with strong domination of one party. 90% and 95% intervals are displayed.

Table 1: Descriptive statistics

	2011	2016	2021
Panel A: Electoral Setting			
Districts	2,176	1,960	2,072
Polling Sites (schools)	4,924	5,234	11,841
Polling-stations (classrooms)	107,436	77,290	86,482
Classrooms per school	21.8	14.8	7.3
Registered voters per classroom	185.7	296.5	292.3
Panel B: First Round Results			
Participation			
Votes per polling-station	155.5	243.9	207.6
Turnout (%)	84.0	82.1	71.1
Vote (%)			
Winner party	27.8	17.3	15.4
Loser party	20.7	32.7	10.9
Other parties	39.3	32.6	55.0
Blank	8.8	11.9	12.4
Null	3.4	5.5	6.3
Panel C: Second Round (Runoff) Results			
Participation			
Votes per polling-station	153.3	237.3	218.0
Turnout (%)	82.9	80.4	75.1
Vote (%)			
Vote margin	2.7	0.2	0.2
Winner party	48.2	46.9	46.9
Loser party	45.5	46.6	46.6
Blank	0.7	0.8	0.6
Null	5.6	5.7	5.9
Poll-watcher presence			
None	46.4	63.3	52.7
Winner party alone	3.8	5.6	5.1
Loser party alone	8.4	2.8	12.9
Both	41.4	28.2	29.3
Duration (minutes)			
Vote duration	470.6	467.0	700.3
Vote count duration	33.5	40.9	40.7

Note: This table shows descriptive statistics for the first and second (runoff) rounds of the three presidential elections studied in this paper (2011, 2016, and 2021). Panel A shows the aggregate and average characteristics of the electoral setting. Panel B shows results for the first round. In none of the three elections did any party obtain more than 50% of the votes, which triggered a runoff election. Winner and Loser parties refer to the eventual winner and runner-up of the second round. Turnout is computed as the share of registered voters who cast a ballot. Vote shares are expressed as a percentage of total votes cast, including blank and null votes. Panel C shows results for the second round. The vote margin is defined as the difference in vote share between the winning and losing party, with margins of approximately 0.2% in the 2016 and 2021 elections. Pollwatcher presence categories are mutually exclusive and reported as a percentage of classrooms. Vote duration refers to the total number of minutes the polling station was open for voting. Vote count duration refers to the number of minutes taken to count the ballots after closing.

Table 2: Effect of poll-watchers on the vote margin

	Vote margin in percentage points (winner - loser)					
	(1)	(2)	(3)	(4)	(5)	(6)
Winner only	10.75***	0.95***	0.48***	0.41***	0.41***	0.31***
	(0.57)	(0.12)	(0.09)	(0.08)	(0.08)	(0.06)
Loser only	-18.67***	-1.11***	-0.31***	-0.38***	-0.38***	-0.38***
	(0.79)	(0.17)	(0.11)	(0.09)	(0.09)	(0.07)
Both	-3.21***	0.22***	0.10*	-0.00	-0.01	-0.01
	(0.35)	(0.08)	(0.06)	(0.05)	(0.05)	(0.04)
School FE	No	No	Yes	Yes	Yes	Yes
District FE	No	Yes	No	No	No	No
First round controls	No	No	No	Yes	Yes	Yes
Poll-workers demographics	No	No	No	No	Yes	No
Mean	1.39	1.39	1.39	1.39	1.39	2.14
N	170,412	170,412	170,412	170,412	170,412	258,288

Note: This table reports estimates of the effect of poll-watcher presence on the second round vote margin, expressed in percentage points as the difference in vote shares between the winner and runner-up. The three poll-watcher presence categories are mutually exclusive: *Winner only* indicates that only the winning party had a poll-watcher present, *Loser only* indicates that only the losing party had a poll-watcher present, and *Both* indicates that both parties had a poll-watcher present. The omitted category is polling stations with no poll-watcher present. Columns (1) through (5) report results for a common subsample with complete observations for all variables, for comparability. Column (1) reports the baseline estimate with no controls or fixed effects. Column (2) adds district fixed effects. Column (3) adds polling-site (school) fixed effects, which captures most of the variation in poll-watcher assignment and its relationship with vote margins. Column (4) further adds a predicted vote margin based on first round results (see equation (2)). Column (5) adds poll worker demographics, including a gender indicator proxied by first name and an above-30 age indicator proxied by national ID number. Column (6) shows the preferred specification, which uses the full sample and excludes the demographic controls from column (5). Standard errors are clustered at the polling-site level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table 3: Effect of poll-watchers on vote shares, turnout, vote duration, and count duration

	Vote shares (percentage points)					Participation, and voting and count duration		
	(1) Margin (Winner – Loser)	(2) Winner	(3) Loser	(4) Null	(5) Blank	(6) Turnout	(7) Voting duration (min)	(8) Count duration (min)
Winner only	0.31***	0.16***	-0.16***	0.01	-0.01**	-0.00	-0.21	0.13
	(0.06)	(0.04)	(0.04)	(0.06)	(0.01)	(0.03)	(0.38)	(0.22)
Loser only	-0.38***	-0.12**	0.23***	-0.11	0.00	-0.02	-0.90*	-0.56*
	(0.07)	(0.05)	(0.05)	(0.07)	(0.01)	(0.04)	(0.49)	(0.30)
Both	-0.01	0.05*	0.03	-0.08**	0.00	0.00	-0.41*	1.27***
	(0.04)	(0.03)	(0.03)	(0.04)	(0.00)	(0.02)	(0.24)	(0.15)
School FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First round controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean	2.14	47.82	45.68	5.76	0.74	80.79	537.09	38.26
N	258,288	258,288	258,288	258,288	258,288	258,289	199,342	206,570

Note: This table reports estimates of the effect of poll-watcher presence on second round outcomes at the polling station level. The three poll-watcher categories are mutually exclusive: *Winner only* indicates that only the winning party had a poll-watcher present, *Loser only* indicates that only the losing party had a poll-watcher present, and *Both* indicates that both parties had a poll-watcher present. The omitted category is polling stations with no poll-watcher present. Columns (1) through (5) report effects on vote shares, expressed as a percentage of total votes cast. Column (1) reports the effect on the margin between the winner and loser. Columns (2) and (3) report effects on the winner and loser vote shares respectively. Columns (4) and (5) report effects on null and blank vote shares. Column (6) reports the effect on turnout, measured as the share of registered voters who cast a ballot. All share variables are expressed in percentage points (0 to 100). Columns (7) and (8) report effects on voting duration and count duration in minutes respectively. The smaller sample in these columns reflects missing duration data for some polling stations due to difficulties extracting this information from the voting registries. All specifications include polling-site fixed effects and first round controls, defined as the predicted second round vote share based on first round results (see equation (2)). Standard errors are clustered at the polling-site level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Appendix

A1 Data Collection

To construct the poll-watcher presence data, I used a combination of direct data access, image-to-text software, manual review, and AI image recognition. The steps followed were: (i) accessing vote counts from the first and second rounds directly through the ONPE database, along with the 271,208 scanned voting registry records for each round, totaling 542,416 records across the two rounds of the three elections containing poll-watcher information; (ii) using Layout-Parser (Shen et al. (2021)), an image-to-text software, to analyze the handwritten sections of each registry and assign poll-watcher status to each classroom, this software identifies text blocks as shown in Figure A.1, but because it is designed for printed rather than handwritten text, recognition is imperfect; (iii) since only one poll-watcher per party is permitted, cases where two poll-watcher text blocks are detected are coded as having both parties present, even when party affiliation cannot be independently confirmed for each; (iv) once a preliminary assignment to each poll-watcher presence category is determined, I proceed to manually review the registries to correct any misclassifications;[19] and (v) all registries are then processed through AI image recognition[20] to more precisely identify timestamps[21],

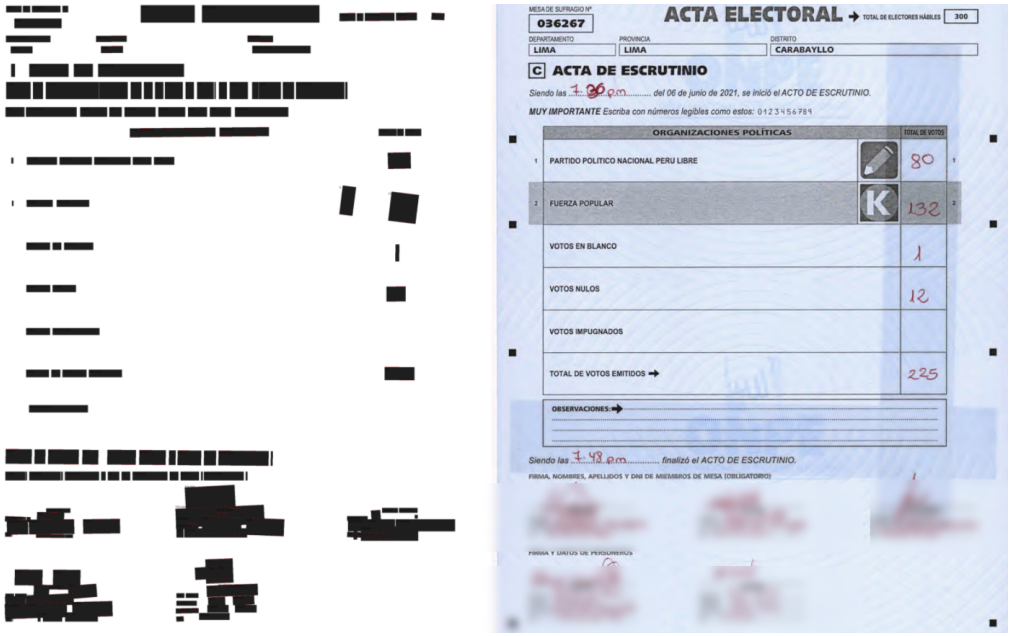
[19]The manual review was conducted separately for each pre-assigned group, which expedited the process since most entries were already correct.

[20]Cropped images for each timestamp and each poll-worker and poll-watcher entry were analyzed using Gemini 2.5 Flash Lite.

[21]To validate the accuracy of the extracted timestamps, I manually reviewed a random sample of 30 classrooms per year for each of the four time variables (classroom opening, closing, and vote count start and end times). Overall, 5.5% of extracted timestamps do not match the manually verified values. Error rates are substantially higher in 2011 (12.1%) than in 2016 (2.1%) and 2021 (2.1%), likely reflecting lower document quality in the earlier election. A large share of errors in 2011 correspond to cases where the extraction algorithm returned a missing value despite a legible timestamp being present in the image. Restricting the sample to observations where the algorithm did return a value, the error rate falls to 2.4% overall, 5.5% in 2011, 0.4% in 2016, and 1.3% in 2021, indicating that conditional on extraction, the algorithm reads timestamps accurately. These error rates are small and unlikely to meaningfully affect my estimates.

national ID numbers, and full names of poll-workers and poll-watchers, as well as the party affiliation of poll-workers, which is also used to validate and further improve poll-watcher identification.

Figure A.1: Voting registry and algorithm to detect handwritten text



Notes: The figure on the left shows how the Layout-Parser software identifies textblocks from the print and handwritten information in the right. Personal information from poll-workers and poll-watchers has been blurred for this example.

A2 Other Figures and Tables

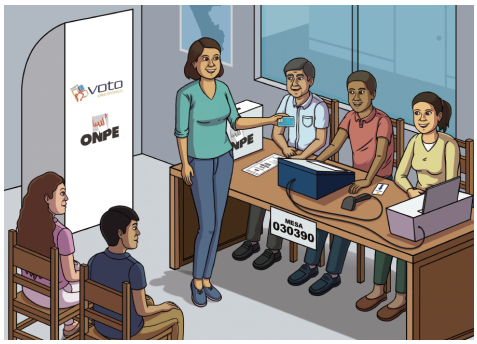
Figure B.1: Election day in Peru



(a) Electoral site (school)



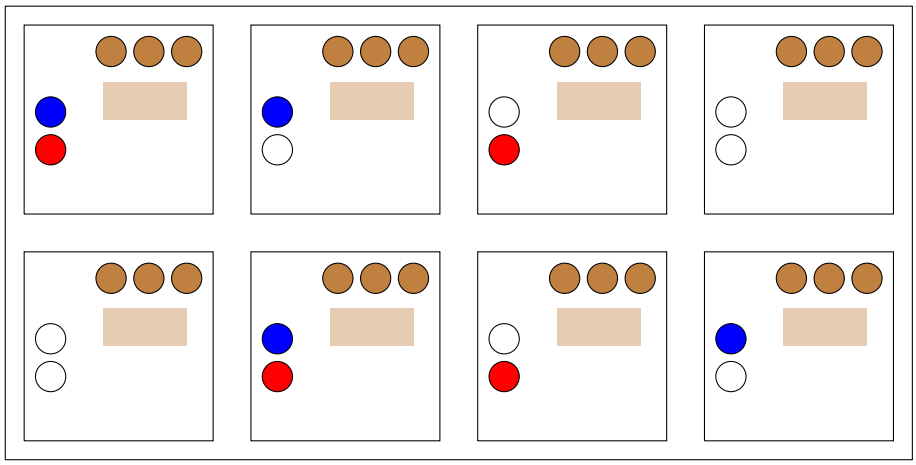
(b) Line outside electoral table (classroom)



(c) Table members (right), voter (center) and observers (left) in a voting table (classroom)

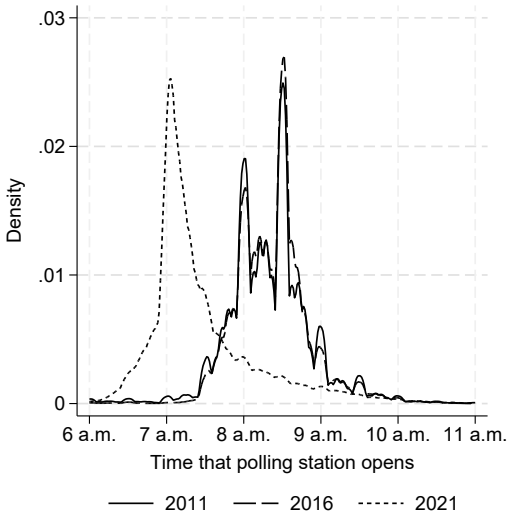
Notes: These images show how voting takes place on election day. Image (a) shows the outside of a polling site (school), where voters can see which classroom they are assigned to based on their ID and last name. Image (b) shows voters in a polling station (classroom) lined up and waiting for their turn to vote. Image (c) shows the different actors inside the classroom: three poll-workers (top right), drawn from the voter pool assigned to that classroom, are responsible for overseeing the voting process, verifying IDs, and counting votes. The voter (middle) steps behind a voting booth to cast their ballot. The party poll-watchers (bottom left) monitor the process and may flag any irregularities. During the vote count, they may challenge any ballot they believe was counted improperly.

Figure B.2: Research Design

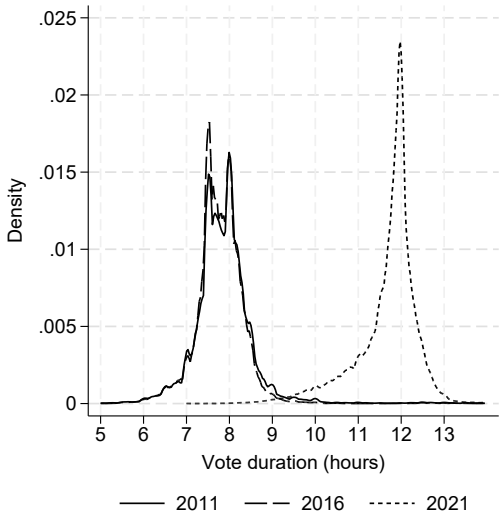


Notes: The following figures shows an example of an election site with 8 tables. Each of these has one of 4 possible observer assignments (red and blue). The analysis focuses on how each of the 4 possible cases deviates from the site mean considering that once controlling for first round votes, within a site, there is not much more than parties can do to inform their allocation that would be related to potential outcomes.

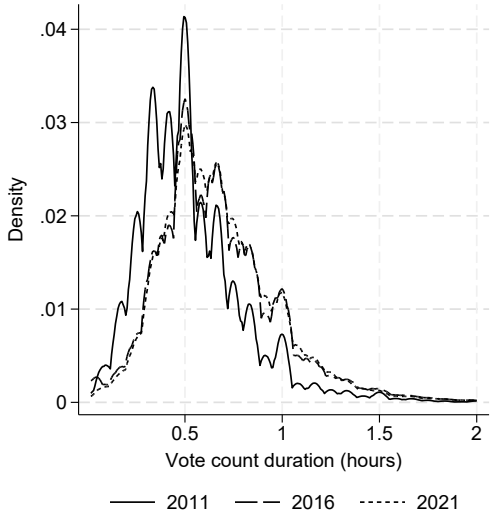
Figure B.3: Distribution of polling station opening times, vote duration, and vote count duration.



(a) Time polling station opens



(b) Voting duration



(c) Vote count duration

Notes: Panel (a) shows the distribution of polling station opening times across election years. Official opening time is 8:00 a.m. in 2011 and 2016, and 7:00 a.m. in 2021. In 2011, 29.4% of stations opened on time, with 92.3% open within one hour and 99.2% within two hours. In 2016, 24.3% opened on time, 92.6% within one hour, and 99.5% within two hours. In 2021, 23.1% opened on time, 83.5% within one hour, and 95.2% within two hours. Panel (b) shows the distribution of total voting duration. Polls were scheduled to close at 4:00 p.m. in 2011 and 2016, and at 7:00 p.m. in 2021; the extended closing time in 2021, introduced due to the COVID-19 pandemic, explains the longer voting durations observed that year. Panel (c) shows the distribution of vote-counting duration.